

RADHIKA KUMBHAR

Data Scientist | Python, SQL | Machine Learning & GenAI

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Data Scientist and Entry-Level AI/ML Engineer with hands-on experience in data analytics, machine learning, deep learning, and GenAI systems. Skilled in Python, advanced SQL, statistics, and building end-to-end AI solutions including predictive models, computer vision (ViT, GANs), and Retrieval-Augmented Generation (RAG) applications.

TECHNICAL SKILLS

Programming & Querying: Python (Advanced), SQL (Advanced), Java (Basic)

Data Analytics: EDA, Data Cleaning, FE, Statistical Analysis, Hypothesis Testing, Visualization

Machine Learning: Regression, Classification, Clustering, Random Forest, XGBoost, Model Evaluation

Deep Learning: CNNs, Vision Transformer (ViT), GANs

GenAI : RAG, Embeddings, Vector Search, LangChain, LangGraph

Libraries & Tools: Pandas, NumPy, Scikit-learn, Matplotlib, PyTorch, Pydantic, Streamlit, Flask, Power BI

Databases & Platforms: MySQL, PostgreSQL, Pinecone/FAISS (Basic), Git, Docker (Basic), Jupyter, Kaggle, Google Colab

EXPERIENCE

Ignnitte Center of Excellence (COE) — Trainer & Developer, Industry 4.0 | Aug 2025 – Present

- Delivered hands-on training in ML, AI, IoT, and Industry 4.0 technologies.

EdTech Society, IIT Bombay — Deep Learning Intern | 2023

- Built and trained Vision Transformer (ViT) models for image classification tasks.

PROJECTS

Medical Chatbot using RAG | LangChain, OpenAI, Pinecone, Flask

[\[link\]](#)

- Developed an **AI-powered chatbot** capable of answering medical queries by extracting context-aware information from medical PDFs.
- Integrated **LangChain** with **Pinecone vector database** and **retrieval-augmented generation**.
- Implemented document parsing and chunk-based querying to ensure accurate, real-time responses from large unstructured documents.

SAR Image Colorization (Hybrid cGAN) | PyTorch, CNN, Vision Transformer

- Designed and implemented a hybrid **conditional Generative Adversarial Network (cGAN)** architecture that combines **Convolutional Neural Networks (CNN)** and **Vision Transformers (ViT)** to colorize grayscale Synthetic Aperture Radar (SAR) images.
- Integrated the CNN encoder for spatial feature extraction and a ViT-based decoder for capturing long-range dependencies and global context during color generation.
- Evaluated model performance using image quality metrics such as **Structural Similarity Index Measure (SSIM)** and **Peak Signal-to-Noise Ratio (PSNR)**.

EDUCATION

B.Tech in Computer Science & Engineering

KIT's College of Engineering, Kolhapur | 2021 – 2025 | CGPA : 8.00/10